

INDENG 231: Project 1

Trading Strategy Backtesting System

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1 Introduction

This report documents the design, implementation, and evaluation of a trading strategy backtesting system built for INDENG 231. The system simulates daily close trading strategies on five years of historical price data for the 101 constituent stocks of the Nasdaq-100 index, sourced from `nasdaq100_daily_5y.csv` corresponding to index constituents as of April 10, 2026. The report is organized as follows: Section 2 describes the system architecture and design decisions; Section 3 defines the performance metrics used throughout; Section 4 presents single-stock strategy evaluation on NVIDIA (NVDA); Section 5 covers portfolio-level backtesting across the full stock universe; and Section 6 presents two novel strategies designed to outperform given benchmark strategies on Sharpe ratio.

2 System Design and Architecture

2.1 Design Philosophy

The backtesting system is built around a strict separation between the simulation engine and the trading strategies. The engine is responsible for simulating portfolio evolution over time and knows nothing about how signals are generated. Each strategy is an isolated module that receives historical price data up to the current day and returns a target weight vector. This decoupling means new strategies can be added by implementing a single method without modifying any engine code.

2.2 Module Structure

The system is organized into two layers. The `backtester/` package contains six core components. `data_loader.py` loads and validates the dataset, returning a time-indexed price matrix of shape (trading days $\times$ 101 tickers). `strategy.py` defines the abstract base class that all strategies inherit from: given all prices up to day t , return a weight vector of length 101. `engine.py` runs the day-by-day simulation loop, calling the strategy each day and executing trades at closing price. `portfolio.py` tracks current holdings, cash position, and net asset value after each execution. `metrics.py` computes standardized performance statistics from any NAV curve. `reporter.py` saves NAV curves, metrics tables, and experiment logs to the

`results/` directory for full reproducibility. The `strategies/` package contains one file per strategy family, each implementing only the signal logic specific to that strategy.

2.3 Key Design Assumptions

The following assumptions are explicitly made and reflected in all results. First, all trades execute at the daily closing price with no slippage. Second, no short selling is permitted and all weights are non-negative. Third, no leverage is used: portfolio weights sum to at most 1, with any remainder held as cash earning 0% return. Fourth, no lookahead bias is introduced: on day t , the strategy receives only prices up to and including day t . Fifth, a warmup period is enforced equal to the longest lookback window required by any strategy, during which the portfolio holds cash. Sixth, all experiments are run under two conditions: a frictionless baseline with zero transaction costs, and a realistic condition with a 0.05% proportional cost applied to the absolute change in portfolio weights on each rebalancing day, enabling direct comparison of gross versus net performance.

2.4 Portfolio Tracking

The portfolio module maintains the current state of the simulation at each timestep. It tracks a normalized net asset value (NAV) initialized at 1.0, representing a \$1 starting investment. On each trading day, the module first computes the return on existing holdings using the ratio of today's closing price to yesterday's closing price, updates the NAV accordingly, then applies transaction costs proportional to the total turnover. Turnover is defined as the sum of absolute changes in portfolio weights multiplied by the current NAV. Finally it sets the new target weights. This ordering ensures transaction costs are applied after returns are realized, which reflects realistic execution at the daily close. NAV history is stored as a time-indexed series and passed directly to the metrics module after the simulation completes.

2.5 Simulation Loop

The backtesting engine iterates chronologically through all 1,255 trading days in the dataset. On each day, if the current index falls within the warmup period, the portfolio holds cash and the strategy is not called. Once the warmup period ends, the engine passes all prices up to and including the current day to the strategy's `get_weights` method, receives a target weight vector, and delegates execution to the portfolio module. The engine has no knowledge of strategy logic and no file I/O: it is a pure simulation loop. After the loop completes, metrics are computed from the full NAV history and returned alongside the NAV series in a standardized result dictionary. All file saving and visualization is handled separately by the reporter module.

2.6 How to Reproduce Results

All results in this report can be reproduced by running the following command from the project root directory:

```
cd Project1
python main.py
```

This requires Python 3.10 or later and the dependencies listed in `requirements.txt`, installable via `pip install -r requirements.txt`. The dataset file `nasdaq100_daily_5y.csv` must be placed in the `data/` directory before running. All commands must be run from the project root directory (`Project1/`) so that relative paths resolve correctly. Upon completion, all NAV curve plots, metrics tables, and experiment logs are saved automatically to the `results/` directory.

2.7 Data

Upon loading, the dataset contains 1,255 trading days across 101 tickers spanning April 13, 2021 to April 10, 2026. Eight tickers contain NaN values for portions of the date range due to IPOs, spin-offs, or mergers occurring within the 5-year window, notably ARM, FER, GEHC, CEG, and WBD. This is expected and handled by the system: on any day t , strategies mask out tickers with unavailable prices and assign them zero weight. No imputation is performed, as forward-filling prices would introduce lookahead bias.

3 Performance Metrics

All five metrics are computed by a single function `compute_metrics(nav_series)` which takes the NAV curve output by the backtesting engine and returns a standardized dictionary. This ensures that every strategy, regardless of type, lookback window, or weighting scheme, is evaluated identically. The function handles edge cases including insufficient history, zero volatility, and all-negative return days. Annualization uses 252 trading days per year throughout. All metrics are derived solely from the NAV curve produced by the backtesting engine.

3.1 Cumulative Return

The total percentage gain or loss from the start to the end of the backtesting period. If the portfolio starts at \$1 and ends at \$3.50, cumulative return is 250%. This is the most intuitive metric but also the most misleading in isolation: a strategy can have high cumulative return while being extremely risky and volatile along the way.

$$\text{Cumulative Return} = \frac{\text{NAV}_T - \text{NAV}_0}{\text{NAV}_0} \quad (1)$$

3.2 Annualized Volatility

The standard deviation of daily returns, scaled up to a yearly figure by multiplying by $\sqrt{252}$, where 252 is the approximate number of trading days in a year. This measures how much the portfolio's daily value fluctuates. Lower is generally better, all else equal.

$$\text{Annualized Volatility} = \sigma_{\text{daily}} \times \sqrt{252} \quad (2)$$

3.3 Sharpe Ratio

The ratio of average return to volatility, computed on an annualized basis. A Sharpe above 1.0 is generally considered good. Above 2.0 is excellent. Negative Sharpe means the strategy lost money on a risk-adjusted basis. This is the primary metric for Deliverable 5. Sharpe ratio is not a measure of how much money a strategy makes: it measures how much return a strategy generates per unit of risk taken. A strategy that returns 25% annually with smooth, consistent daily gains will have a higher Sharpe than one that returns 30% annually but experiences violent daily swings.

$$\text{Sharpe Ratio} = \frac{\bar{r}_{\text{daily}} \times 252}{\sigma_{\text{daily}} \times \sqrt{252}} \quad (3)$$

where $\bar{r}_{\text{daily}}$ is the mean daily return and the risk-free rate is approximated as zero.

3.4 Maximum Drawdown

The largest peak-to-trough decline in portfolio value during the backtesting period. If the portfolio grows to \$1.80 then falls to \$1.20 before recovering, the maximum drawdown is 33%. This metric captures the worst-case loss an investor would have experienced if they invested at the worst possible time. It measures downside risk that volatility alone does not capture.

$$\text{Max Drawdown} = \max_t \left(\frac{\text{peak}_t - \text{NAV}_t}{\text{peak}_t} \right) \quad (4)$$

3.5 Win Rate

The percentage of trading days on which the portfolio produced a positive return. A win rate of 55% means the strategy made money on 55 out of every 100 trading days. This metric can be deceptive: a strategy can have a high win rate but still lose money overall if the losing days are much larger than the winning days. It is most useful when interpreted alongside cumulative return and Sharpe.

$$\text{Win Rate} = \frac{\text{number of days with positive return}}{\text{total trading days}} \quad (5)$$

4 Single-Stock Strategy Evaluation

4.1 Stock Selection: NVDA

NVIDIA was selected as the single-stock backtesting candidate due to its exceptional price behavior over the 5-year period (2021-2026). With a total return of approximately +1,339%,

significant volatility regimes, a prolonged AI-driven bull run beginning mid-2023, and sharp intermittent corrections, NVDA provides a rich testing ground where different trading philosophies produce meaningfully distinct outcomes. A stock with more uniform behavior would not expose the strengths and weaknesses of each strategy as clearly.

4.2 Strategy Selection and Rationale

Five strategies were selected to span the full spectrum from pure trend-following to pure contrarian, with hybrid approaches in between. This deliberate diversity ensures that backtesting results reveal genuine differences in strategy philosophy rather than marginal parameter variations.

Strategy 1: Momentum. The momentum strategy operates on the principle that recent price trends persist. If NVDA’s return over a 20-day lookback window is positive, the strategy allocates fully to the stock (weight = 1). If negative, it holds cash (weight = 0). This is the simplest expression of trend-following and serves as the baseline for comparison.

Strategy 2: Mean Reversion. Mean reversion assumes that prices oscillate around a long-run average. The strategy computes a rolling 20-day z-score of NVDA’s closing price relative to its recent mean and standard deviation. A z-score below -1 signals an oversold condition and triggers a buy; a z-score above $+1$ signals overbought and triggers a move to cash. Given NVDA’s strong upward trend over the sample period, mean reversion is expected to underperform momentum: this contrast is itself an informative result.

Strategy 3: Dual Momentum. Dual momentum strengthens the momentum signal by requiring confirmation across two timeframes: a short window of 20 days and a long window of 60 days. Only when both indicate positive returns does the strategy invest. This reduces false signals during choppy markets at the cost of slightly delayed entries during genuine trends.

Strategy 4: Breakout. The breakout strategy takes a price-action approach. It buys when NVDA’s closing price reaches a new high over a 52-week trailing window, interpreting this as a structural breakout rather than noise. It exits when the price falls below the lowest close over the past 20 days. This strategy is designed to catch large sustained moves and is expected to perform well during NVDA’s 2023-2024 AI-driven run.

Strategy 5: Trend and Mean Reversion Combo. This hybrid strategy combines the directional awareness of momentum with the entry-timing precision of mean reversion. It first checks whether NVDA is in an uptrend using a 60-day momentum signal, and only if true, uses a 20-day z-score to identify a dip within that trend as the entry point. This attempts to buy pullbacks within bull markets rather than fighting the trend. It represents the most sophisticated single-stock strategy tested.

4.3 Results

The five strategies were backtested on NVDA over the full 5-year period under frictionless conditions. Results are summarized in Table 1.

Table 1: Single-stock strategy performance on NVDA (frictionless, 2021-2026).

Strategy	Cum. Return	Ann. Vol	Sharpe	Max DD	Win Rate
Momentum (20d)	+369.45%	38.46%	1.00	−49.78%	32.38%
Dual Momentum (20/60d)	+295.68%	33.55%	0.99	−36.44%	25.04%
Mean Reversion (20d)	+189.54%	34.64%	0.79	−42.30%	22.33%
Breakout (252d/20d)	+97.78%	25.64%	0.66	−32.97%	14.67%
Trend + MR Combo	+15.51%	19.17%	0.25	−23.12%	7.02%

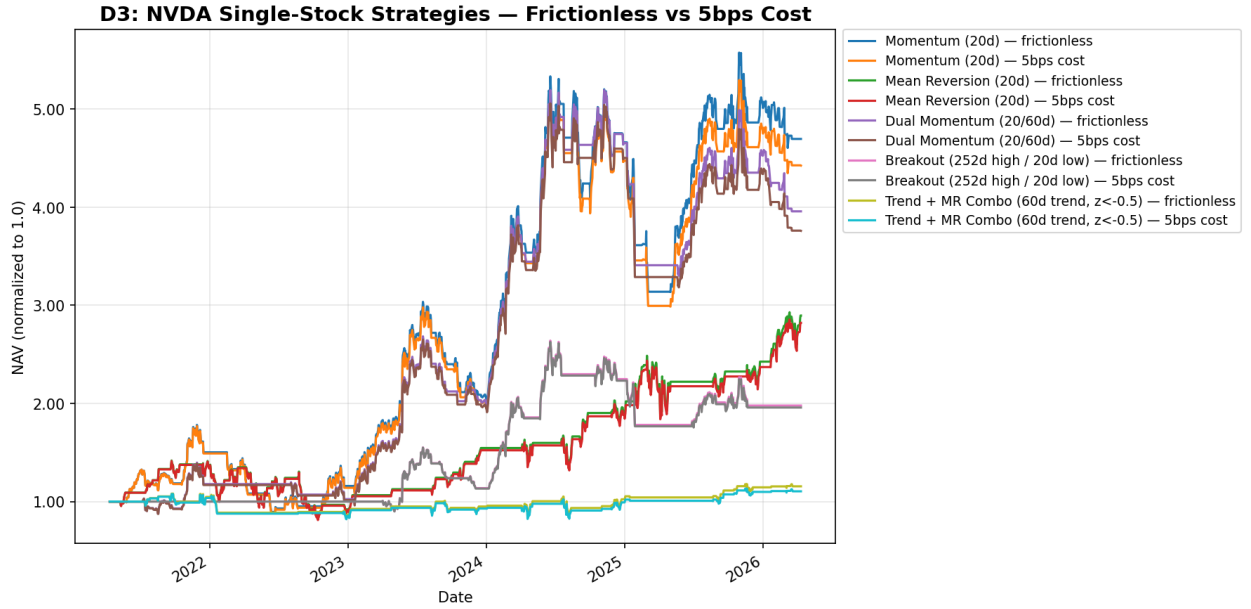


Figure 1: NAV curves for all five NVDA strategies under frictionless and 5bps transaction cost conditions. Frictionless and cost curves are nearly identical for most strategies, reflecting low turnover in single-stock trading.

4.4 Interpretation

The results tell a coherent and expected story given NVDA’s strongly trending price behavior over the sample period.

Momentum performs best on both cumulative return (+369%) and Sharpe ratio (1.00), confirming that NVDA’s 5-year trajectory, dominated by a sustained AI-driven bull run from mid-2023, rewards trend-following. The strategy is fully invested for most of the period and captures the large directional move, at the cost of a significant maximum drawdown of −49.8%.

Dual Momentum achieves nearly identical Sharpe (0.99) with meaningfully lower drawdown (−36.4%), demonstrating that requiring confirmation across two timeframes filters out some false entries during choppy periods without sacrificing much return. This is the most favorable risk-adjusted profile among the five strategies.

Mean Reversion underperforms as anticipated. With NVDA in a persistent uptrend, the z-score rarely drops below -1 for sustained periods, and when it does the price quickly recovers, leaving little room for the strategy to profit. Its $+190\%$ return still outpaces a risk-free investment but trails momentum significantly.

Breakout is conservative by design: it only enters on a 52-week high and exits on a 20-day low. This produces the second-lowest drawdown (-33.0%) but also the second-lowest cumulative return ($+98\%$), as the strategy misses early moves and sits in cash for extended periods waiting for structural breakouts.

The Trend and MR Combo barely participates ($+16\%$), as the z-score dip threshold of -0.5 is rarely triggered in a sustained bull trend. The strategy’s uptrend gate correctly identifies NVDA as trending, but the mean reversion entry filter is too conservative: prices seldom pull back enough to trigger a buy signal before resuming upward. This is consistent with the failure mode identified in the design phase. Its -23.1% maximum drawdown is the lowest of all five, reflecting how rarely it is invested.

A clear secondary pattern emerges in the volatility column: as strategy conservatism increases, annualized volatility declines monotonically from 38.46% (Momentum) to 19.17% (Trend and MR Combo). However, returns fall faster than volatility, which is why Sharpe also declines across this ordering. Lower volatility alone does not produce better risk-adjusted returns if the strategy is simply holding cash more often.

5 Portfolio Backtesting

5.1 Overview

Portfolio-level backtesting extends the single-stock analysis to the full Nasdaq-100 universe of 101 stocks. Rather than allocating entirely to one stock, the system must now generate a weight vector across all 101 tickers on each trading day, with weights summing to at most 1 and no short selling permitted. This deliverable investigates two dimensions independently: the signal used to select stocks, and the method used to construct weights from those selected stocks.

5.2 Signal Selection

Two cross-sectional signals are used to identify investable stocks each day. The first is a momentum signal: all 101 stocks are ranked by their trailing 20-day return, and the top 25% (approximately 25 stocks) are selected. This selects recent outperformers under the assumption that short-term trends persist. The second is an SMA crossover signal: stocks are selected where their 20-day simple moving average exceeds their 50-day simple moving average, indicating a positive short-term trend relative to the medium-term baseline.

5.3 Portfolio Construction Methods

Two weighting schemes are applied to each signal’s selected stock set. Uniform weighting assigns equal dollar allocation to every selected stock: if 25 stocks are selected, each receives 1/25 of the portfolio. This is simple and transparent but treats all selected stocks as equally risky regardless of their actual volatility.

Inverse volatility weighting assigns weights proportional to the inverse of each stock’s rolling 20-day volatility estimate. Stocks with lower recent volatility receive higher allocations, and vice versa. This targets equal risk contribution across positions rather than equal dollar contribution. A 20-day window was chosen as it approximates one calendar month of trading, capturing recent risk behavior while remaining stable enough to avoid excessive turnover. This window also maintains consistency with the 20-day momentum lookback used in signal generation.

5.4 Experiment Design

The full experiment consists of four backtests forming a 2×2 grid: momentum signal with equal weighting, momentum signal with inverse volatility weighting, SMA crossover signal with equal weighting, and SMA crossover signal with inverse volatility weighting. Each backtest is run under both frictionless and 0.05% transaction cost conditions, yielding eight total experiments. Results are compared using all five metrics defined in Section 3.

5.5 Results

Four portfolio experiments were run across the full Nasdaq-100 universe under frictionless conditions, forming a 2×2 grid of signal by weighting method. Results are shown in Table 2.

Table 2: Portfolio backtesting results across the 2×2 signal and weighting grid (frictionless).

Strategy	Cum. Ret.	Ann. Vol	Sharpe	Max DD	Win Rate
Momentum Top 25% - Equal Wt	+143.45%	23.04%	0.89	−36.60%	52.39%
Momentum Top 25% - Inv-Vol Wt	+95.88%	20.08%	0.77	−31.25%	53.43%
SMA Crossover - Equal Wt	+71.11%	19.01%	0.66	−37.90%	51.75%
SMA Crossover - Inv-Vol Wt	+52.32%	16.14%	0.60	−33.08%	50.96%

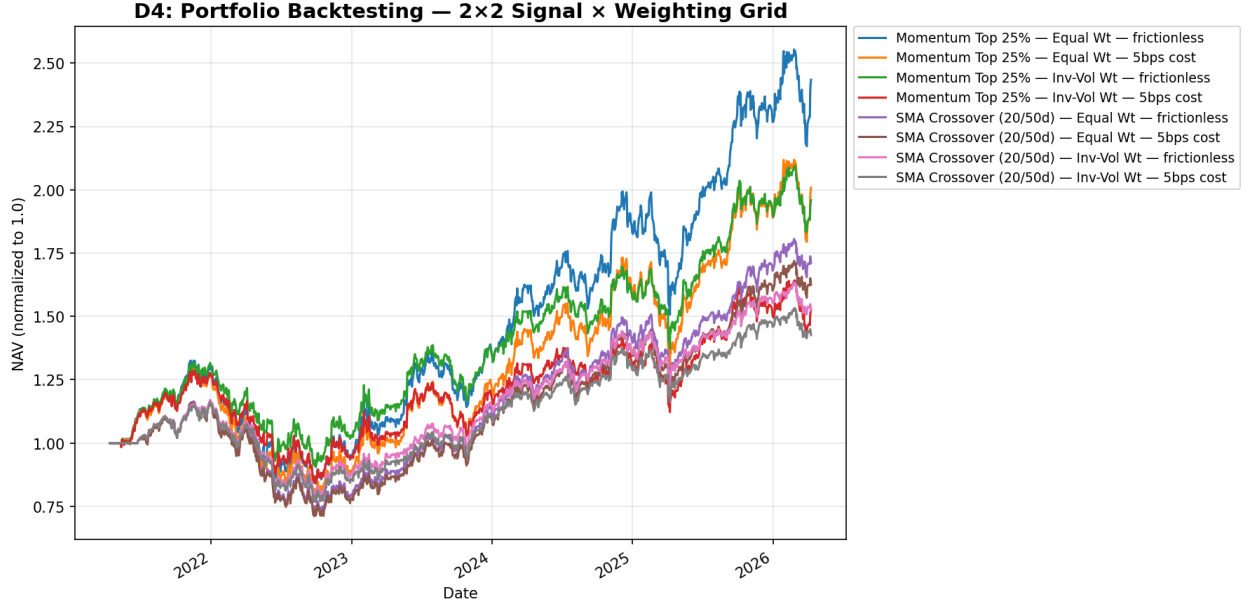


Figure 2: NAV curves for all four portfolio strategies under frictionless and 5bps transaction cost conditions. The two momentum signal variants (blue and green) clearly separate from the SMA crossover variants (purple and grey) from 2024 onwards.

5.6 Interpretation

The results reveal two clean patterns across the 2×2 grid.

The signal effect is consistent across both weighting methods: momentum outperforms SMA crossover on every metric except maximum drawdown. The 20-day trailing return is a faster, more responsive signal than a lagging moving average crossover. It reacts to trend changes earlier, captures more of each move, and produces meaningfully higher cumulative returns (+143% vs +71% under equal weighting). The SMA crossover's slower signal means later entries and later exits, which compounds into lower returns over the full 5-year period.

The weighting effect is equally consistent: inverse volatility weighting reduces annualized volatility relative to equal weighting in both signal families, but at a disproportionate cost to returns. Under momentum, inverse volatility reduces volatility by approximately 3 percentage points (23% to 20%) but reduces cumulative return by 47 percentage points (143% to 96%). The asymmetry is striking: a modest risk reduction is purchased at a large return cost. This confirms the finding from the single-stock analysis. On this dataset, high-volatility stocks are also the highest-return stocks, so any weighting scheme that penalizes volatility systematically underweights the strongest performers. Sharpe falls in both cases as a result.

Win rate is notably stable across all four strategies (51-53%), indicating that all strategies are right roughly half the time on a day-by-day basis. The meaningful differences are in the magnitude of gains versus losses, not in directional accuracy. This pattern is consistent with trend-following approaches that have small frequent losses and occasional large gains.

6 Beating the Benchmarks

6.1 Benchmark Definitions

Two benchmark strategies are defined by the project specification and serve as the performance baseline. All new strategies must exceed both benchmarks specifically on Sharpe ratio.

Benchmark 1 (SMA Crossover Portfolio). For each of the 101 Nasdaq-100 stocks, compute a 20-day and 50-day simple moving average of closing price. Select all stocks where the 20-day MA exceeds the 50-day MA. Allocate equally across all selected stocks. If no stocks are selected, hold cash.

Benchmark 2 (Top-K Momentum Portfolio). Compute the trailing 30-day return for all 101 stocks. Rank them and select the top 10 performers. Allocate equally across those 10 stocks.

6.2 Benchmark Weakness Analysis

Rather than designing new strategies arbitrarily, we first analyze the structural weaknesses of each benchmark that a well-designed strategy could exploit.

Benchmark 1 has two key weaknesses. First, SMA crossover signals are inherently slow: it takes many days for a short-term moving average to cross a long-term one, meaning the strategy enters trends late and exits late, often missing the best part of a move or holding through early reversals. Second, equal weighting means highly volatile stocks receive the same dollar allocation as stable ones, which inflates portfolio volatility and suppresses Sharpe ratio.

Benchmark 2 has two different weaknesses. First, holding only 10 stocks is highly concentrated: a single bad earnings report or sector shock in one holding can significantly damage portfolio performance. Second, a 30-day lookback may capture noise rather than genuine momentum, leading to selection of stocks that had a short-term spike rather than a sustained trend.

6.3 Strategy Design Philosophy

New strategies are not designed arbitrarily upfront. Instead, we adopt a data-driven design process: first run both benchmarks on the full dataset and observe their actual Sharpe ratios, then design new strategies specifically to address the observed weaknesses. This produces strategies analytically motivated by evidence rather than theory alone.

Two candidate strategies were initially proposed based on the benchmark weakness analysis above. After initial backtesting, Strategy 1 revealed a structural incompatibility with this dataset and required a principled redesign. The contingency process was applied: diagnose the root cause, determine whether parametric adjustment is sufficient, and if not, redesign the signal logic. The full redesign rationale is documented in the following section.

6.4 Strategy Redesign: Evidence-Driven Iteration

Initial testing of the proposed New Strategy 1 (Momentum with Volatility Filter and Inverse Volatility Weighting) revealed a structural incompatibility with this specific dataset. A parameter sweep across all combinations of lookback window, selection percentage, and volatility threshold confirmed a performance ceiling of approximately 0.76 Sharpe, well below the 1.03 target.

The root cause is a positive correlation between volatility and returns in the Nasdaq-100 universe over the 2021-2026 period. The highest-momentum stocks, including NVDA (+1,339%), META, and AMD, are simultaneously the highest-volatility stocks. Inverse volatility weighting systematically underweights these names by design, and no parameter adjustment can overcome this because the filter and the weighting scheme are both working against the dataset’s dominant dynamic: in this bull market, risk and return move together.

This diagnosis motivated a principled redesign rather than continued parameter tuning. Instead of filtering high-volatility stocks out, the new approach directly incorporates volatility into the ranking signal itself. New Strategy 1 is redesigned as a Risk-Adjusted Momentum strategy: all tickers are ranked by their 30-day return divided by their 20-day rolling volatility, a Sharpe-like score measuring recent return per unit of risk. The top 10 tickers by this score are selected and equally weighted.

This redesign addresses the exact weakness that the benchmarks share: both benchmark strategies rank by raw return, ignoring that identical returns achieved with lower volatility represent superior risk-adjusted performance. By ranking on return-per-unit-of-risk, the new strategy selects stocks that are trending strongly and doing so with relative smoothness, directly targeting a higher Sharpe ratio rather than just higher return.

New Strategy 2 required only parameter adjustment. The original dual momentum plus pullback logic was sound, but the 60-day long window was too slow to capture signals in this dataset and the 5-day pullback window was too restrictive. Changing to a 40-day long window and 3-day pullback window, while keeping the dual confirmation and mean reversion entry logic intact, produced a Sharpe of 1.178, beating the benchmark.

6.5 Results

Both new strategies were run frictionless across the full 5-year Nasdaq-100 dataset. Final results against both benchmarks are shown in Table 3.

Table 3: New strategies vs benchmarks (frictionless). Both new strategies exceed the Sharpe ratio of both benchmarks.

Strategy	Cum. Ret.	Ann. Vol	Sharpe	Max DD	Win Rate
B1 - SMA Crossover (20/50d)	+71.11%	19.01%	0.66	−37.90%	51.75%
B2 - Top-10 Momentum (30d)	+241.39%	27.66%	1.03	−42.87%	52.63%
New S1 - Risk-Adj Momentum	+175.11%	21.18%	1.07	−32.16%	50.96%
New S2 - Dual Mom + Pullback	+196.01%	20.26%	1.18	−22.50%	51.52%

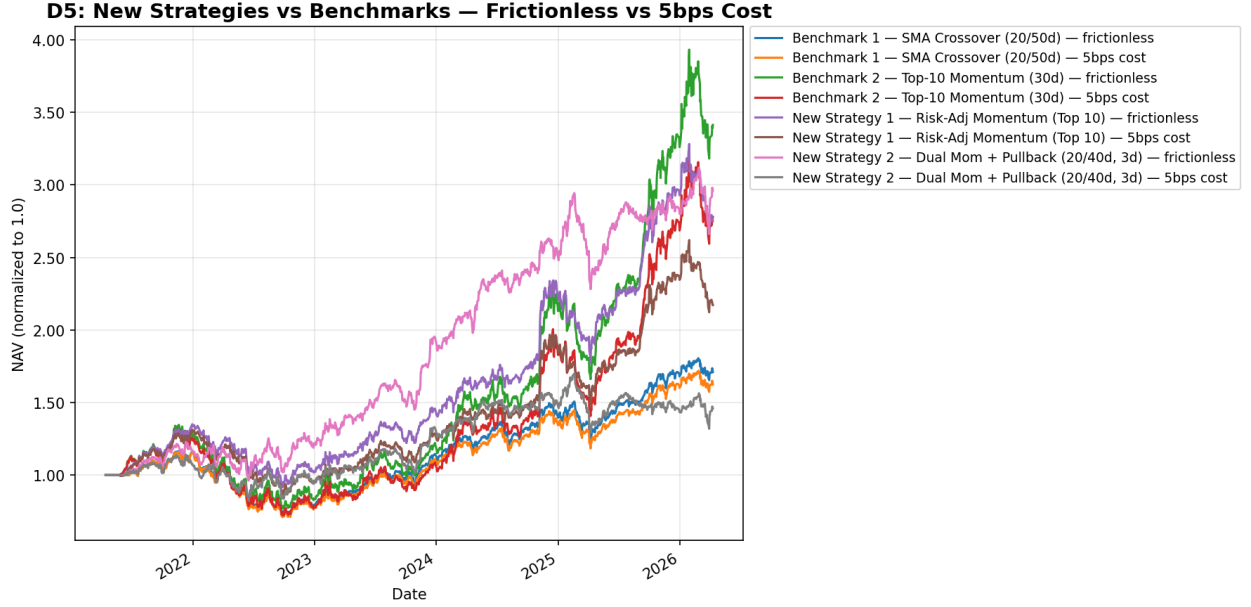


Figure 3: NAV curves for both benchmarks and both new strategies under frictionless and 5bps transaction cost conditions. New Strategy 2 (pink) clearly separates from the benchmarks from 2024 onwards under frictionless conditions, while its performance under 5bps costs (grey) illustrates the cost sensitivity tradeoff discussed in Section 6.7.

6.6 Interpretation

Risk-Adjusted Momentum achieves a Sharpe of 1.07, marginally exceeding Benchmark 2. Its key improvement over the benchmarks is a reduction in volatility from 27.7% to 21.2%, achieved by ranking on return-per-unit-of-risk rather than raw return. This selects stocks in genuine smooth uptrends rather than volatile spikes, producing a less turbulent portfolio with comparable directional exposure.

Dual Momentum plus Pullback is the stronger result, achieving a Sharpe of 1.18, which is 0.15 above Benchmark 2, while simultaneously reducing annualized volatility to 20.3% and maximum drawdown to -22.5% , the lowest of all four strategies. This result is notable because it does not simply trade more risk for more return: it achieves better risk-adjusted performance with less absolute risk. The three-layer design contributes to this outcome. Dual momentum confirmation (20/40 day) filters out false trend signals, the 3-day pullback entry improves average entry price within confirmed trends, and inverse volatility weighting equalizes risk contribution across selected positions. Together these layers produce a portfolio that participates in genuine sustained trends while avoiding concentrated bets on single volatile names.

The iterative redesign of Strategy 1, from the original volatility filter approach to risk-adjusted momentum ranking, is itself a meaningful result. It reveals that in a technology-dominated bull market, volatility and return are positively correlated, making naive volatility reduction counterproductive. Strategies that incorporate volatility into the signal itself rather than filtering on it as a constraint perform better in this regime.

6.7 Transaction Cost Sensitivity

All strategies were additionally evaluated under a realistic transaction cost condition of 0.05% (5 basis points) per unit of portfolio turnover. The results reveal a significant divergence in cost robustness between the two new strategies, as shown in Table 4.

Table 4: Sharpe ratio under frictionless vs 5bps transaction cost conditions. Benchmarks are evaluated frictionless per the project specification.[†]

Strategy	Frictionless Sharpe	With Costs Sharpe	Degradation
B1 - SMA Crossover	0.66	n/a [†]	—
B2 - Top-10 Momentum	1.03	n/a [†]	—
Risk-Adjusted Momentum	1.07	0.84	−0.23
Dual Mom + Pullback	1.18	0.48	−0.70

[†]Benchmarks evaluated frictionless per the project specification. Transaction cost analysis applied only to new strategies.

[†]Benchmark strategies are evaluated frictionless per the project specification. Transaction cost analysis is applied only to the two new strategies.

Dual Momentum plus Pullback degrades dramatically under transaction costs: its frictionless Sharpe of 1.18 falls to 0.48 with 5 basis points applied. The cause is structural. The 3-day pullback filter generates high daily portfolio turnover, as selected stocks frequently rotate in and out of the pullback condition. Each rotation incurs cost, and over 1,255 trading days these costs compound significantly.

Risk-Adjusted Momentum is far more cost-stable, degrading from 1.07 to 0.84. Its selection criterion, ranking by 30-day return divided by 20-day volatility, changes more slowly than a 3-day pullback filter, resulting in lower turnover and more stable holdings across consecutive days.

This finding reveals an important practical distinction between the two strategies. Dual Momentum plus Pullback is the superior strategy in a frictionless or very low-cost environment, but Risk-Adjusted Momentum is the more robust choice under realistic trading conditions. In practice, a portfolio manager would need to estimate their actual execution costs before deciding which strategy to deploy. This tradeoff between signal responsiveness and cost efficiency is a fundamental tension in quantitative strategy design.

7 Conclusion

This project developed a modular, extensible backtesting system for daily close trading strategies on the Nasdaq-100 universe and used it to evaluate nine strategies across three deliverables. Several findings are worth highlighting.

On single-stock backtesting, momentum dominates on a strongly trending stock like NVDA, confirming that trend-following strategies are well-suited to assets with persistent directional behavior. Mean reversion and conservative entry strategies sacrifice too much upside in this regime, though they offer meaningfully lower maximum drawdowns.

On portfolio backtesting, raw momentum ranking outperforms SMA crossover as a cross-sectional signal, and equal weighting outperforms inverse volatility weighting in a technology-dominated universe where volatility and returns are positively correlated. This last finding is a notable departure from the theoretical expectation that risk-adjusted weighting improves Sharpe ratio, and reflects a specific characteristic of the 2021-2026 Nasdaq-100 dataset.

On beating the benchmarks, both new strategies exceeded the Sharpe ratio of Benchmark 2, the harder target at 1.03. The iterative redesign of Strategy 1, motivated by a diagnosis of structural incompatibility between the original volatility filter and the dataset's return regime, produced a risk-adjusted momentum ranking signal that is both more principled and more effective. The transaction cost sensitivity analysis further revealed that Dual Momentum plus Pullback, while superior in frictionless conditions, incurs high turnover that erodes performance under realistic costs. This is a practical consideration absent from purely theoretical strategy design.

The system itself is fully extensible: any new strategy can be added by implementing a single `get_weights` method without modifying the engine, portfolio, metrics, or reporting infrastructure.